

# CHRISTIAN KEOUGH

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## EDUCATION

### Bachelor of Science in Mechanical Engineering

May 2028

Minor in Computer Science

GPA: 3.86

University of Florida, Gainesville, FL

Relevant Coursework: Statics, Computer Aided Graphics and Design, Thermodynamics, Physics with Calc 2, MATLAB, Materials

### Associate in Engineering

April 2024

Indian River State College, Fort Pierce, FL

GPA: 3.93

Honors: Valedictorian, Class of 800 students (2024); Dean's List (2023-Present); AICE Diploma with Merit

## PROFESSIONAL EXPERIENCE

### Florida Advanced Manufacturing and Systems Integration Laboratory

Undergraduate Lab Assistant

March 2026 – Present

- Programming G-code + Hansen Solubility interface to reduce RED calc time by 95% for DIW VIPS multi-layer lattice printing.
- Collaborating with PhD mentor and lab team to design a manifold controlling RH and Z-axis variability in VIPS DIW.
- Conducting literature reviews of 15+ VIPS papers to guide biomaterial fabrication, presenting progress in monthly seminars.

### PGA Homes and Roofing

Roofing Assistant

Summer 2024 & Summer 2025

- Installed and reinforced tile roof systems using bull tar, membrane, polyurethane foam; improved durability of 20 roof sections.
- Reinforced weak points on multi-story roofs by transporting and preparing concrete with power mixer; handled ~15 loads.
- Applied liquid silicone coatings to waterproof susceptible areas, covering 2,000-3,000 sq ft and extending roof lifespan.

## INVOLVEMENT

### Generational Relief in Prosthetics (GRiP)

Production Design Team and 3D Printing Subteam Member

Spring 2026 – Present

- Addressing stability needs for child bike pedal prosthetic in SolidWorks, enabling effective prototyping and TPU/PLA 3D printing; collaborating closely with team members by attending 13 total meetings.
- Extracting 15+ critical dimensions via ImageJ to support CAD modeling and prosthetic design accuracy in SolidWorks.

### Design Build Fly (DBF)

Structures Subteam Member

Fall 2025

- Modeled aileron skin geometry in SolidWorks; iterated ~5 designs, collaborated with wing subteam for fit + manufacturability.
- Assisted in fabrication preparation and maintained an organized workshop environment.

### FIRST Robotics Competition (FRC), VEX Robotics Program

Mechanical Lead / Vice President

Fall 2020 – Spring 2022

- Coordinated design and assembly of VEX robots, integrated sensors, drivetrain across 2 robot iterations (autonomous + driver).

### Judo Club

Member

Fall 2025

- Trained 2 to 4 hours weekly, practicing live sparring (randori); earned Yellow Belt rank.

## PROJECTS

### Canine Pelvic Brace with Hip Socket CAD Design

Spring 2026

- Fabricated CAD canine pelvic brace from CT scans via 3D splines, anatomical planes; captured ~25 anatomical references.
- Integrated hip socket assistive device with joint articulation and structural analysis using Abaqus; ran 3 simulation validations.

### Hansen Solubility Material Selection Interface

Spring 2026 – Present

- Developed Python tool parsing CSV solvent data, analyzing ~30 solvents, and generating 3D/2D compatibility plots.
- Applied Hansen solubility interface for DIW VIPS solvent selection; screened 30 comparisons for most optimal pairing.

### Adjustable Chair Design

Fall 2025

- Designed adjustable chair with knee-sitting and leaning positions with ergonomic CAD, constraints, and screw alignment.

## SKILLS & CERTIFICATIONS

Skills: SolidWorks CAD Design Associate (CSWA), Autodesk Inventor (2022), Onshape, 3D Printing, FEA, C++, Python, MATLAB